

 DEPARTMENT OF MECHANICAL ENGINEERING UNIVERSITY OF MORATUWA, MORATUWA.			MODULE INFORMATION 24th June – 4th November
Course	B.Sc. Engineering (Hons.)	Session	Semester 5 (21 st Intake)
Module Code	ME 3813	Title	Machine Design
Credits	02	Hours / Week	1 Lecture, 2/1 Tutorials
Module Coordinator	Dr. Eranga De Silva (EDS) [MC, L] ahteranga@uom.lk		
Resource Personnel	Prof. Nirosh Jayaweera (NJ) [L] nirosh@uom.lk		
	Dr. Pasan Henadeera (PH) [L] henadeerapb@uom.lk		
	Mr. Prabhathiya Herath (PR) [L] herathrp@uom.lk		
	Mrs. Ransini Gunaratne (RG) [L] ransinih@uom.lk		
Instructors	Mr. Dilka Abeynayake (AA) [I] Mr. Kavishka Kariyawasam (BB) [I] Mr. Pulasthi Dabare (CC) [I]		
L – Lecturer, MC – Module Coordinator, I – Instructor			
Learning outcomes: After completing this course module, the students should be able to • innovatively think and look at a machine or a design with an inquiring mind and convert an idea or a concept to a practical design • carry out a design task with confidence • communicate design information through technical reports, engineering drawings, oral communication, etc. • work as a team to manage a design project			

DESIGN PARAMETER SHEET FOR MACHINE DESIGN PROJECT

Group No	Vehicle Parameters		Stages of gearbox	
	Maximum Speed (km/h)	Vehicle Mass (kg)	Forward Gears	Reverse Gears
1	210	2150	5	1
2	210	1800	5	1
3	130	1700	5	1
4	150	2250	5	1
5	110	1250	5	1
6	190	1900	5	1
7	180	1600	5	1
8	100	2100	5	1
9	180	2000	5	1
10	170	1800	5	1
11	170	2050	5	1
12	210	1700	5	1
13	130	1900	5	1
14	170	2200	5	1
15	120	2100	5	1

The car should be able to travel on a road inclined at **30 degrees** to the horizontal at a speed of **35 km/h**.

The Group Numbers are as follows:

Group Number	Group Name	Index	Name
1	2JZ	210458X	PERERA G.D.D.C.
	2JZ	210678V	WANNIARACHCHI S.C.P.
	2JZ	210681A	WARUSHAVITHANA S.S.
2	Cheron Alpha	200121B	DHANANJAYA K.D.L.
	Cheron Alpha	210357L	MALAGE T.R.
	Cheron Alpha	210700J	WICKRAMAGE W.S.V.L.
3	Gear Mavericks	210101A	DE SILVA G.A.I.
	Gear Mavericks	210347G	MADHUBHASHINEE H.A.W.R.
	Gear Mavericks	210361T	MALLAWA ARACHCHI J.B.
4	Gear Pro	210061A	BANDARA H.M.K.D.
	Gear Pro	210308N	KULATUNGA B.N.
	Gear Pro	210410U	NAVINNA R.R.
5	GearTech Trio	210042R	ANJANA E.A.O.
	GearTech Trio	210525C	RANAWEERA RKP
	GearTech Trio	210564T	SANDARU H.W.P.
6	GearUp	210080F	CHATHUMINDA G.A.Y.
	GearUp	210295T	KODITHUWAKKU K.A.S.S.
	GearUp	210345A	LIYANAGE S.W.
7	GearWizards	210143D	DISSANAYAKE D.M.M.C.
	GearWizards	210318U	KUMARASINGHE K.K.A.S.M.
	GearWizards	210319A	KUMARASINGHE K.U.I.
8	N2S	210322C	KUMARASIRI H.S.D.
	N2S	210378C	MAPALAGAMA O.N.
	N2S	210533A	RATHNAMALALA T.N.S.
9	Team Gearsmiths	210016R	ABHIRAMI T
	Team Gearsmiths	210094C	DANINDI H.M.T.
	Team Gearsmiths	210381E	MAYOORATHAN K.
10	Team SKI	210281X	KAUSHIKA K.K.G.
	Team SKI	210447M	PATHIRAGE S.S.K.
	Team SKI	210689F	WEERASINGHE M.G.I.U.
11	TechGear WKS	210494D	PREMAKUMARA H.P.S
	TechGear WKS	210640A	THEMIYA K.L.
	TechGear WKS	210660J	UDAYAKANTHA D.A.W.I.
12	Tensor Gear	210114P	DESHAPRIYA M.P.P.
	Tensor Gear	210311R	KUMARA R.P.D.R.
	Tensor Gear	210722D	WIJESOORIYA W.A.A.D.
13	The Gearheads	210070B	BIBULEWELA P.A.C.
	The Gearheads	210127H	DILSHAN A.K.R.
	The Gearheads	210633F	THARINDU N.L.A.K.

14	Torque Titans	210091N	DAHANAYAKE A.H.
	Torque Titans	210276L	KARUNARATHNE W.K.S.A.
	Torque Titans	210713C	WIJEKOON S.S.J.
15	TriMechanix	210121J	DHARMADASA W.H.M.A.D.
	TriMechanix	210496K	PREMARATHNE O.D.
	TriMechanix	210509G	RAJAPAKSHA W.R.A.K.H.